10.1 Final Examination

104.6/100 Points

3/16/2025

Attempt 1

Review Feedback
3/16/2025

Attempt 1 Score:
104.6/100

Add Comment

Anonymous Grading: **no**

5 Attempts Allowed
3/10/2025 to 3/16/2025

Details

FINAL EXAM

This course’s final assignment is an examination, designed to assess the new data science skills you have gained over the past nine modules. Before beginning this exam, please ensure you have access to cloud computing resources provided in the course. If you have any questions about accessing programming exercises for this course, please review **Python Resources (https://canvas.uw.edu/courses/1780179/pages/g08353e47df56e0495d455cf3a715c629)** .

You are receiving this assignment at the beginning of the module in order to allow you the most amount of time to complete it. During this final module you will still work through new material which may be relevant to this assignment, so bear this in mind as you proceed.

Before you begin, please read the following instructions carefully. Please note that these instructions are different from the problem set instructions you have seen previously.

- **Collaboration is not allowed on this exam. You may only speak with your instructor about this material. As a reminder you can schedule time with your instructor here (https://calendly.com/sophinliu/imt-573-win-2025-final-exam-1-1) .**
- Be sure to include well-documented (e.g., commented) code chunks, figures, and clearly written text chunk explanations as necessary. Any figures should be clearly labeled and appropriately referenced within the text. Be sure that each visualization adds value to your written explanation; avoid redundancy—you do not need four different visualizations of the same pattern/data.
- You must appropriately reference all materials and resources you use within your exam.
- Remember, partial credit will be awarded for each question for which a serious attempt at finding an answer has been shown. Students are **strongly** encouraged to attempt each question and document their reasoning process even if they cannot find the correct answer.
- At the end of the exam you will be asked to sign a **Statement of Compliance** to accompany your submission. The Compliance Statement is found below. You must include this text, word-for-word, in your final exam submission. By including the Statement of Compliance in your final exam submission, you indicate that you have read the statement and agree to its terms. **Failure to do so will result in your exam not being accepted.**

- **You final examination, in notebook form, should be turned in on Canvas.**

Statement of Compliance

I affirm that I have had no conversation regarding this exam with any persons other than the instructor. Further, I certify that the attached work represents my own thinking. Any information, concepts, or words that originate from other sources are cited in accordance with University of Washington guidelines as published in the Academic Code (available on the course website). I am aware of the serious consequences that result from improper discussions with others or from the improper citation of work that is not my own.

Final Exam Notebook: **IMT_573_final_exam_WIN_2025-Your_Name.ipynb (https://canvas.uw.edu/courses/1780179/files/131852807?wrap=1) (https://canvas.uw.edu/courses/1780179/files/131852807/download?download_frd=1)**

Dataset for the final exam: **sales.csv (https://canvas.uw.edu/courses/1780179/files/131832153?wrap=1) (https://canvas.uw.edu/courses/1780179/files/131832153/download?download_frd=1)**

View Rubric

Select Grader

Sophin Liu (Teacher)

IMT 573 Final Exam (1)

Criteria	Ratings		Pts
Problem 1 (a)	**2 pts** **Full Marks**	**0 pts** **No Marks**	2 / 2 pts
Problem 1(b)	**2 pts** **Full Marks**	**0 pts** **No Marks**	2 / 2 pts
Problem 1(c)	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts
	Comments would be beneficial to provide more interpretation of the model coefficients (e.g., how each significant variable impacts the likelihood of having an affair) as well as a more concise thought process.		
Problem 1(d)	**4 pts** **Full Marks**	**0 pts** **No Marks**	4 / 4 pts
	Comments add more context in your explanations. You mention the marriage rating as having the highest negative correlation, but it's important to also mention that this means marriage rating is negatively associated with having an affair, and the odds ratio decreases as marriage rating increases.		
Problem 1(e)	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts

IMT 573 Final Exam (1)

Criteria	Ratings		Pts
Problem 1(f)	**4 pts** **Full Marks** **Comments** -0.1: The stepwise selection process is explained well , but explanation could use more detail on how the final model from this question is different from the one in question 1(c), especially regarding which variables were included or excluded.	**0 pts** **No Marks**	3.9 / 4 pts
Problem 1(g)	**2 pts** **Full Marks**	**0 pts** **No Marks**	2 / 2 pts
Problem 2(a)	**5 pts** **Full Marks** **Comments** Points cutting waived due to package installation issues	**0 pts** **No Marks**	5 / 5 pts
Problem 2(b)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 2(c)	**5 pts** **Full Marks** **Comments** Thorough exploration and analysis!	**0 pts** **No Marks**	5 / 5 pts
Problem 2(d)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 3(a)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 3(b)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 3(c)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 3(d)	**5 pts** **Full Marks** **Comments** -0.1: The explanation is missing details on the actual equation. The intercept (465.87) represents baseline sales when cost is zero, while the slope (0.31) shows that for each 1-unit increase in cost, sales increase by 0.31 units. Great work otherwise!	**0 pts** **No Marks**	4.9 / 5 pts
Problem 4(a)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 4(b)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts

IMT 573 Final Exam (1)

Criteria	Ratings		Pts
Problem 4(c)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
Problem 4(d)	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 / 5 pts
	Comments excellent work going above the requirement for this and comparing with the different models!		
Problem 5(a)	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts
Problem 5(b)	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts
Problem 5(c)	**4 pts** **Full Marks**	**0 pts** **No Marks**	3.8 / 4 pts
	Comments -0.2: could use a clearer comparison of flexible and less flexible models. Adding more detail on when each is best suited, along with examples, would strengthen this answer.		
Problem 6(a) view longer description	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts
Problem 6(b)	**4 pts** **Full Marks**	**0 pts** **No Marks**	4 / 4 pts
Problem 6(c)	**3 pts** **Full Marks**	**0 pts** **No Marks**	3 / 3 pts
Problem 7 (Extra Credit: 5) view longer description	**0 pts** **Full Marks**	**0 pts** **No Marks**	3 / 0 pts
	Comments missing key mathematical calculations but credits given for applying concept. Please note, this question - What is the likelihood that a sample number of tails appears in n tosses , assuming the coin is fair - is more appropriate based on 'Binomial Distribution' . You can approximate 'Poisson distribution' on a binomial distribution when the number of trials is very large and the probability of success (in your case, getting tails) is small.		
Problem 8 (Extra Credit: 5) view longer description	**0 pts** **Full Marks**	**0 pts** **No Marks**	2 / 0 pts
	Comments Points for effort		

Total Points: 104.6

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IMT_573_f...350.ipynb	2.25 MB

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